

LOYALTY SIGNALING AS REPARATIVE STRATEGY

A Text-Mining Analysis of HK CEs' Political Discourse, 1997–2026

Authors: Deng Xinran, Zhong Jingquan, Liu Jing Yi

The University of Hong Kong
Faculty of Social Science

INTRODUCTION

Since the 2020 National Security Law (NSL) and the 2021 electoral overhaul, Hong Kong's public discourse has seen a significant increase in references to the Central People's Government (CPG). This trend is most evident under Chief Executive John Lee, whose administration is often viewed as proactively "Mainlandizing" its rhetoric to signal political alignment. This research investigates the discursive aspect of this phenomenon, exploring how Chief Executive (CE) public addresses resonate with Beijing's language. By quantifying the frequency and style of these signals, we aim to uncover the underlying logic of loyalty signaling in Hong Kong's evolving political landscape. The project seeks to move beyond anecdotal observations by providing a systematic, data-driven analysis of how loyalty is communicated across five administrative terms from 1997 to 2026.

THEORETICAL FRAMEWORK

Drawing upon the theoretical framework of loyalty signaling in authoritarian regimes, this study situates the discourse of Hong Kong Chief Executives (CEs) within the tension between administrative autonomy and political allegiance. While scholarly literature suggests that officials with less-trusted backgrounds or high career dependency on the center have stronger incentives to signal loyalty (Qian & Bai, 2024; Baturo & Khokhlov, 2025), Hong Kong's "One Country, Two Systems" framework initially moderated such behaviors. However, recent scholarship argues that the post-2019 socio-political friction and subsequent electoral overhauls have reconfigured the CE-Central Government relationship into a more traditional hierarchical pattern (Hung, 2020; Kwong, 2025). Addressing the methodological limitations of previous studies that relied on narrow speech samples or arbitrary keyword selection (Peck et al., 2024; Deng & Lin, 2025), this research utilizes a systematic six-dimensional framework and textual mining. It explores whether the intensification of loyalty-signaling rhetoric serves as a strategic, reparative mechanism to restore political trust during periods of institutional uncertainty.

METHODOLOGY

Data sampling and cleaning

The primary dataset comprises 6,994 official press releases and 2,697 speeches/articles in Traditional Chinese, spanning July 1997 to March 2026, all of which scraped from the official websites. This longitudinal corpus covers five CE administrations: Tung Chee-hwa, Donald Tsang, Leung Chun-ying, Carrie Lam, and John Lee. In the meanwhile, to facilitate comparative analysis, central leaders' addresses regarding Hong Kong are collected as a proxy for Mainland political discourse. We conduct rigorous data cleaning (e.g., stripping HTML tags, text normalization) and semantic segmentation using the jieba library. We also implement the TF-IDF algorithm with keyword boosting and customized stop-word filters to isolate loyalty signals from routine administrative language.

Conceptualizing "Loyalty Signaling"

Loyalty signaling is conceptualized into three layers and six categories: Authority signals (Personal Loyalty & Institutional Loyalty), Strategic alignment (Policy Loyalty & Nationalistic Loyalty), and Cognitive convergence (Ideological Loyalty & Rhetorical Loyalty).

Keyword detection

We use a normalized "Loyalty Signaling Keyword Density" metric (hits per 10,000 characters) to quantify how loyalty signaling has fluctuated across different administrations and political climates. Statistical significance is verified using ANOVA and post-hoc comparisons to pinpoint critical inflection points.

Cosine similarity

We quantify the degree of discursive convergence by calculating cosine similarity:

$$\text{cosine similarity} = \frac{A \cdot B}{\|A\| \|B\|}$$

where A and B represent the TF-IDF vectors of CE speeches and central leader addresses respectively, and the metric ranges from 0 (completely dissimilar) to 1 (identical).

LIMITATION & OUTLOOK

Some limitations must be acknowledged. First, the analysis relies exclusively on public speeches and press releases, which may represent a "front-stage" performance that does not fully capture the complexities of private, "back-stage" interactions between SAR officials and Beijing. Second, since we used keyword detection and text similarity as the major methodologies, even though the quantitative metrics provide a robust macro-level overview, they may overlook the subtle linguistic nuances, tone, or coded metaphors that characterize political communication beyond specific keyword densities. Finally, the focus on Chinese-language documents leaves a gap in understanding how the CEs manage multi-audience signaling, particularly when addressing international stakeholders in English.

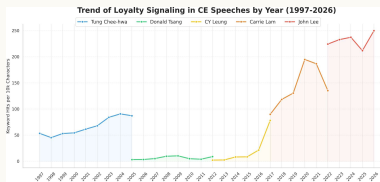
Despite these constraints, this research contributes to the existing literature by enriching the academic understanding of the loyalty-signaling discursive paradigm in semi-autonomous contexts and systematizing the analysis of loyalty-displaying dis-courses in Hong Kong. By developing a six-dimensional keyword framework to concretize "loyalty," this study moves beyond narrating qualitative anecdotes to providing a measurable and empirical basis for analyzing political alignment.

Methodologically, the application of text mining and semantic similarity analysis offers a replicable model for studying political discourse in other Chinese provinces or autonomous regions. Looking forward, this research provides a foundation for future inquiries into the audience reception of modular signaling, particularly among the business sector and international community, regarding whether there are significant audience costs of such high-density loyalty rhetoric. Future studies could also expand this framework to investigate whether the current linguistic balance is a sustainable equilibrium or a transitional phase that will eventually converge with mainland political rhetoric.

FINDINGS & ANALYSIS

Trends of Loyalty Signaling in CE Speeches

The study reveals significant variations in loyalty signaling density across different administrations. While Tung Chee-hwa established a moderate baseline, the subsequent terms of Donald Tsang and Leung Chun-ying saw a substantial decline, with median keyword densities often dropping to zero. This trend was dramatically reversed during Carrie Lam's tenure, where signaling escalated exponentially following the 2017 inauguration and peaked during the 2020 political turmoil. Under John Lee, loyalty signaling reached its zenith in both intensity and consistency, suggesting that this discursive practice has transitioned from a situational response to "trust crises" into a standardized and institutionalized component of the Chief Executive's official rhetoric.



CE	n	Descriptive Statistics			Post-hoc Comparison (p-adj)			
		Mean	Median	CV	Tung	Tsang	Leung	Lam
Tung	497	61.15	50.98	0.84	—	—	—	—
Tsang	396	16.20	0.00	2.51	***	—	—	—
Leung	451	29.41	0.00	1.74	***	.20	—	—
Lam	646	139.93	112.40	0.76	***	***	***	—
Lee	707	228.80	211.60	0.56	***	***	***	***

Statistical Summary and Pairwise Comparison of Loyalty Keyword Density in CE Speeches
Note. Overall ANOVA: $F(4, 2692) = 558.02, p < .001$. *** $p < .001$; ** $p < .01$; * $p < .05$.

Distributional Characteristics of Loyalty Dimensions

In terms of the specific dimensions of loyalty, the data highlights a clear structural preference within the executive discourse. "Ideological loyalty" emerged as the most prevalent category across all administrations, particularly under John Lee and Tung Chee-hwa. Furthermore, CEs consistently exhibited high levels of "Institutional" and "Policy" loyalty, emphasizing their adherence to the national constitutional order and central policy directives. Conversely, there was a notable lack of "Personal loyalty" expressions across the board, and no significant evidence of "Rhetorical loyalty" (the mimicry of Mainland-style bureaucratic jargon). This indicates that signaling is primarily framed as a systemic and institutional alignment rather than an individualized or wholesale linguistic assimilation.

Loyalty Keyword Density Across Chief Executive Tenures (1997-2026)

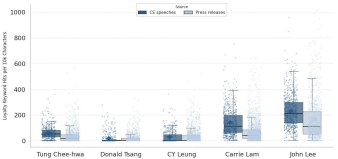
Values represent average occurrences of loyalty signaling terms per 10,000 characters in CE speeches.

	Personal Loyalty	Institutional Loyalty	Policy Loyalty	Ideological Loyalty	Rhetorical Loyalty
Tung Chee-hwa	44.7	4.7	5.1	6.7	0
Donald Tsang	12.8	5.8	0.6	0.5	0.5
Leung Chun-ying	5.9	5.9	3.5	0.3	2.7
Carrie Lam	67.7	18.4	26.9	7.2	22.3
John Lee	115.4	19.1	17.7	8.7	28.4

Channel Divergence and Institutionalization in Press Releases

While Chief Executives Tung Chee-hwa, Carrie Lam, and John Lee significantly prioritized loyalty signaling in their spoken speeches over written press releases, Donald Tsang and Leung Chun-ying showed no such statistical gap, as their overall signaling remained uniformly low across both channels.

Distribution of Loyalty Keyword Density by Channel and Chief Executive (1997-2026)



CE	n(S)	n(P)	Mean(S)	Mean(P)	t-stat	p-value
Tung	499	1338	60.9	34.46	9.60	***
Tsang	423	1105	17.43	17.39	0.02	.99
Leung	451	1603	29.41	31.47	-0.74	.46
Lam	646	1878	139.93	67.74	15.59	***
Lee	707	1070	228.80	165.14	9.02	***

Comparison of Loyalty Keyword Density (Welch's t-test)

Note. S = Speeches, P = Press Releases.

Welch's t-test was used due to unequal variances and sample sizes. *** $p < .001$; ** $p < .01$; * $p < .05$.

We also discover a dynamic shift in communication channels. Until the early years of the Lee administration, loyalty signaling was significantly more concentrated in the CEs' personal speeches. However, in 2026, the signaling density in press releases surpassed that of personal speeches for the first time, which may suggest an institutionalization of such political signaling within the broader civil service.

Textual Similarity and the "Modular" Signaling Strategy

Despite the marked increase in loyalty-related keywords, cosine similarity analysis shows that the overall proximity between CE speeches and central leaders' addresses remains consistently low. Instead of a wholesale linguistic "Mainlandization", the result suggests that CEs employ a "modular" signaling strategy. They strategically integrate high-precision keywords to signal alignment, while simultaneously maintaining the pragmatic and technocratic linguistic characteristic of Hong Kong's governance tradition.

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